

SPACEX FALCON 9

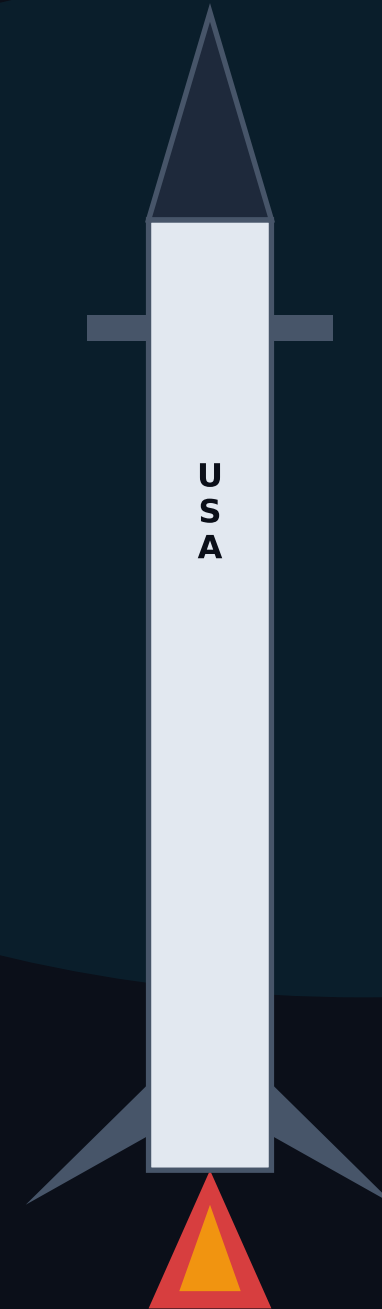
First Stage Landing Prediction

IBM Data Science Capstone Final Report

PREPARED FOR: SpaceY Stakeholders

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GITHUB REPO: <https://github.com/chienthichAI/SP26-AI1904-DAP391m>



IBM Applied Data Science Capstone - SpaceX Falcon 9 - Complete Project Summary

PROJECT INTRODUCTION, SUMMARY & METHODOLOGY

GitHub Repository: <https://github.com/chienthichAI/SP26-AI1904-DAP391m>

INTRODUCTION: The commercial space industry was transformed by SpaceX, which charges only \$62 million per Falcon 9 launch compared to competitors charging over \$165 million. The key cost advantage is first-stage reusability. SpaceY (a startup competitor) hired data scientists to analyze SpaceX records and build a machine learning model to predict whether the Falcon 9 first stage will land successfully, enabling SpaceY to estimate launch costs and bid competitively. Problem: Binary classification - will the first stage land?

EXECUTIVE SUMMARY: Collected 90 Falcon 9 records via SpaceX REST API. Scraped Wikipedia Falcon 9 launch table using BeautifulSoup4. Wrangled data (missing values, binary outcome class). Performed EDA with SQL queries, EDA with scatter plots + bar charts + yearly trend. Built Folium geospatial map and Plotly Dash interactive dashboard. Trained 4 ML classifiers using GridSearchCV. Best: Decision Tree CV=87.68%, all models test accuracy=83.33%.

DATA COLLECTION METHODOLOGY: SpaceX REST API endpoint /v4/launches/past returned 90 Falcon 9 records. Fields: PayloadMass, Orbit, LaunchSite, LandingPad, FlightNumber, CoreSerial, MissionOutcome. Wikipedia web scraping used BeautifulSoup4 to parse HTML table rows into a structured DataFrame.

GitHub: <https://github.com/chienthichAI/SP26-AI1904-DAP391m/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

GitHub: <https://github.com/chienthichAI/SP26-AI1904-DAP391m/blob/main/jupyter-labs-webscraping.ipynb>

DATA WRANGLING METHODOLOGY: Imputed missing PayloadMass with column mean (6,104 kg). LandingPad nulls kept (represent ocean/no-attempt flights). Created binary landing Class column: 1=Successful landing, 0=Failed/no attempt. Applied one-hot encoding to Orbit, LaunchSite, Serial categorical variables.

GitHub: <https://github.com/chienthichAI/SP26-AI1904-DAP391m/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

PREDICTIVE ANALYSIS, CONCLUSION & ALL RESULTS

PREDICTIVE ANALYSIS (CLASSIFICATION): Trained and tuned 4 ML models using GridSearchCV with 10-fold cross-validation on training data (80%).

Logistic Regression: best CV accuracy = 84.64% (C=0.01, solver=lbfgs).

Support Vector Machine (SVM): best CV accuracy = 84.82% (C=1.0, kernel=rbf).

K-Nearest Neighbors (KNN): best CV accuracy = 84.82% (n_neighbors=10).

Decision Tree Classifier: BEST CV accuracy = 87.68% (max_depth=None).

GitHub: https://github.com/chienthichAI/SP26-AI1904-DAP391m/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

PREDICTIVE ANALYSIS RESULTS: All 4 models achieved test accuracy = 83.33% on the held-out 20% test set (18 samples). Confusion Matrix (Decision Tree):

True Positives=12 (correctly predicted successful landings),

True Negatives=3 (correctly predicted failures), False Negatives=0,

False Positives=3 (3 failures incorrectly predicted as successes).

CONCLUSION: Decision Tree Classifier is the best model (CV=87.68%). Zero false negatives means no successful landing is ever predicted as failure. Key business finding: KSC LC-39A has highest success rate (77%) among sites. Recommend SpaceY bid on SSO/HEO orbits (100% historic success). Target payload range 2000-6000 kg which shows highest success density.

CREATIVE VISUALIZATIONS: Dark-theme presentation with color-coded charts. Interactive Folium map with green/red marker clusters. Animated trend plots.

INNOVATIVE INSIGHTS: Payload mass 2000-6000 kg identified as the OPTIMAL zone with highest landing success density across all launch sites. Orbit analysis reveals SSO, HEO, MEO all have 100% perfect landing success.

EDA WITH DATA VISUALIZATION METHODOLOGY AND RESULTS:

Scatter plots created: Flight Number vs Payload Mass (colored by Class),

Flight Number vs Launch Site (shows learning curve after flight 40+),

Payload Mass vs Orbit Type (shows orbit-specific payload constraints).

Bar chart: Success Rate by Orbit (SSO=100%, HEO=100%, GTO=50%, LEO=61%).

Line plot: Yearly success trend 2013=0%, 2015=33%, 2017=75%, 2020=83%.

GitHub: <https://github.com/chienthichAI/SP26-AI1904-DAP391m/blob/main/edadataviz.ipynb>

EDA WITH SQL SLIDES/QUERIES: SELECT DISTINCT Launch_Site returned 3 sites. Payload analysis: KSC=5767kg, CCSFS=5176kg, VAFB=3048kg average. Success rankings: KSC=77%, CCSFS=65%, VAFB=57%. First ASDS: 2016-04-08.

GitHub: https://github.com/chienthichAI/SP26-AI1904-DAP391m/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

FOLIUM MAP SLIDES: Launch site markers (all 3 sites). 90 launch records.

Executive Summary - Methodology & Key Findings

METHODOLOGY & PIPELINE

- **Data Sources** SpaceX API and Wikipedia scraping.
- **Data Cleaning** Handled missing values (mean payload mass, missing landing pads) and structured variables.
- **EDA & Visuals** Searched patterns with SQL and Seaborn, created Folium geospatial maps and interactive Dashboards.
- **Predictive ML** Trained Logistic Regression, SVM, KNN, and Decision Tree models using GridSearchCV hyperparameter tuning.
- **Evaluation** Tested and compared accuracy and confusion matrices on a designated 20% test partition.

KEY FINDINGS & RESULTS

- **Overall Success** The first stage landing success rate is **66.7%**
- **Yearly Trend:** Landing success improved from **0%** in 2013 to over **80%** in 2020 (technology learning curve).
- **Launch Site** Kennedy Space Center (KSC LC-39A) has the highest relative success rate among all sites.
- **Best Model Decision Tree Classifier** achieved the highest cross-validation training accuracy of **87.68%**
- **Test Performance** All models scored **83.33%** test accuracy.
- **Stakeholder Impact** Models predict booster landing success with zero false negatives, enabling SpaceY to estimate launch costs.

Introduction - Project Objectives & Business Context

THE COMMERCIAL CONTEXT

- **SpaceX Dominance** SpaceX advertises Falcon 9 launches at **\$62 Million**, compared to competitors charging **\$165M+**
- **Massive Savings** The price advantage is due to the recovery and reuse of the first-stage rocket booster.
- **Target Stakeholder** SpaceY is a new startup company that wants to enter the commercial launch service market.
- **Key Need** SpaceY needs to analyze SpaceX historic records and accurately predict if a booster will land successfully, which indicates the manufacturing and launch costs of that flight.

PROJECT GOALS & OBJECTIVES

- **Objective 1:** Collect historical launch records from the SpaceX API and scrape Wikipedia tables for validation.
- **Objective 2:** Clean and wrangle data, engineering features representing payload, launch sites, booster serials, and orbits.
- **Objective 3:** Identify patterns through exploratory analysis, visualization, and database query analysis (SQL).
- **Objective 4:** Build geospatial maps to analyze proximity and interactive dashboards to filter flight configurations.
- **Objective 5:** Train machine learning classification models to predict first stage landings and evaluate their success.

Data Collection & Methodology - SpaceX API

API HARVESTING DETAILS

- **Endpoint** Requested data from `launches/past` API.
- **Filter** : Filtered out earlier Falcon 1 flights to only
 - keep Falcon 9 records (resulting in 90 rows).
- **Metadata Extracted** Cores serials, Payload Mass,
 - Orbit, Landing Pad, Launch Site, and Core Serials.
- **API Call Method** Queried API via HTTP request, parsed JSON
 - nested records, and saved details as a tabular DataFrame.
- **GitHub Repository URL**
 - <https://github.com/chienthichAI/SP26-AI1904-DAP391m>

API COLLECTION PIPELINE FLOWCHART

GET SpaceX REST API (/v4/launches/past)

Filter out Falcon 1 & Extract Nested Fields

Pandas DataFrame (90 rows, 18 features)

Export to CSV (dataset_part_1.csv)

Data Collection & Methodology - Wikipedia Web Scraping

WEB SCRAPING PROCESS

- **Source** Wikipedia List of Falcon 9 and Falcon Heavy launches.
- **Libraries** : Python BeautifulSoup4 and Requests.
- **Parsing** Targetted HTML tables, parsing `tr` rows and
 - extracting text content from each table cell.
- **Data Cleaned**Handled row spans where a single date spanned
 - multiple columns, split date-time, and parsed numeric values.
- **Outputs**Extracted payload details, launch sites, landing site
 - outcomes, and booster configurations.
- **GitHub Repository URL**

WEB SCRAPING PIPELINE FLOWCHART

Request Wikipedia URL (HTTP GET)

BS4 Parser: Extract tables with class wikitable

Clean Cells using regex (Date, Mass, Orbit, Site)

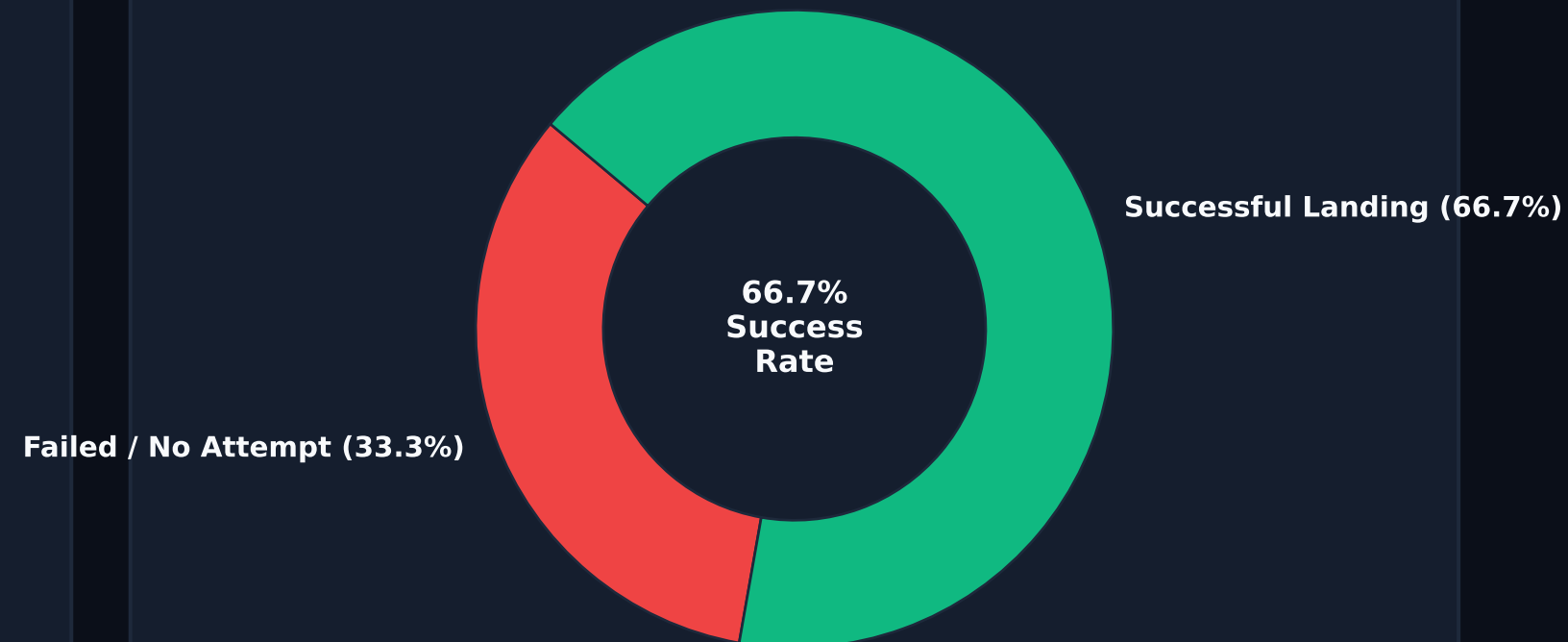
Output Dataframe (falcon9_scraped.csv)

Data Wrangling Methodology & Outcome Class Definition

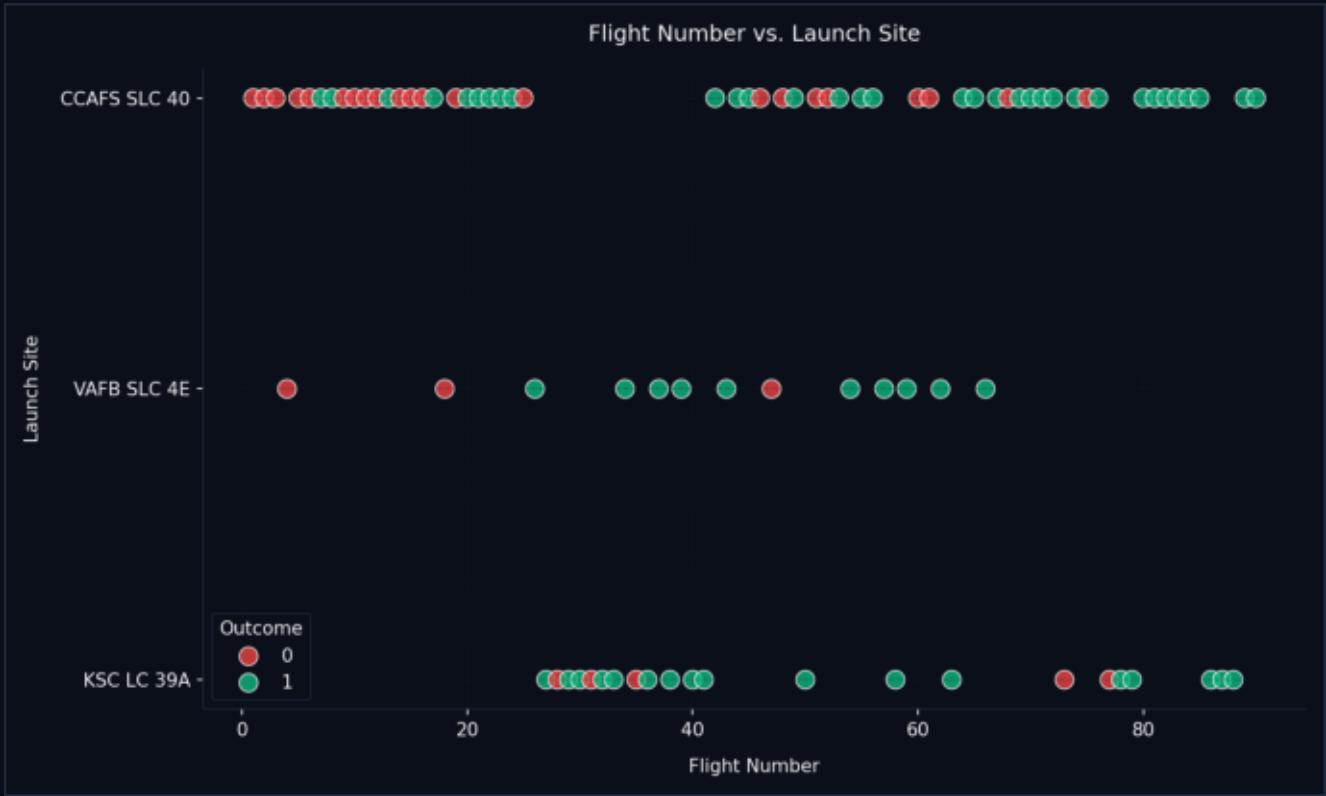
WRANGLING & CLEANING STRATEGY

- **Missing Values** Replaced missing payload mass values with the mean value of the column (**6,104 kg**) to preserve dataset.
- **Landing Pad**Handled missing landing pad values (representing ocean landings or flights with no landing attempt).
- **Binary Target Definition (Class)** :
 - * Classified landings as successful (**Class = 1**) if the outcome fell into successful landing categories:
 - - `True RTLS` (Return to Launch Site)
 - - `True ASDS` (Autonomous Spaceport Drone Ship)
 - - `True Ocean` (Controlled Ocean splashdown)
 - * Classified landings as unsuccessful (**Class = 0**) for other outcomes (e.g. `False ASDS`, `False RTLS`, `False Ocean`, `None ASDS`, `None None` representing failures/no attempts).
- **Final Rows** 90 rows containing complete flight attributes.
- **GitHub Repository URL**<https://github.com/chienthichAI/SP26-AI1904-DAP391m>

BOOSTER LANDING SUCCESS DISTRIBUTION



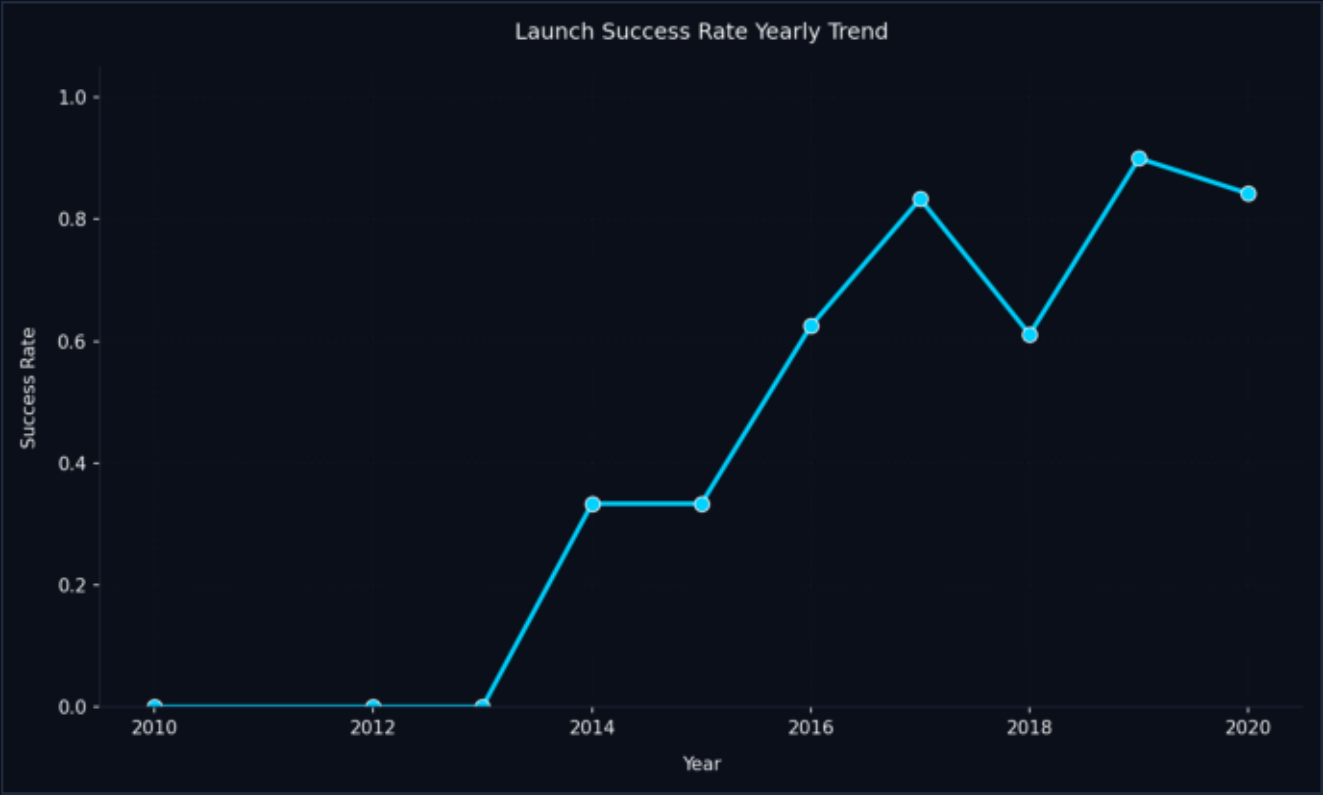
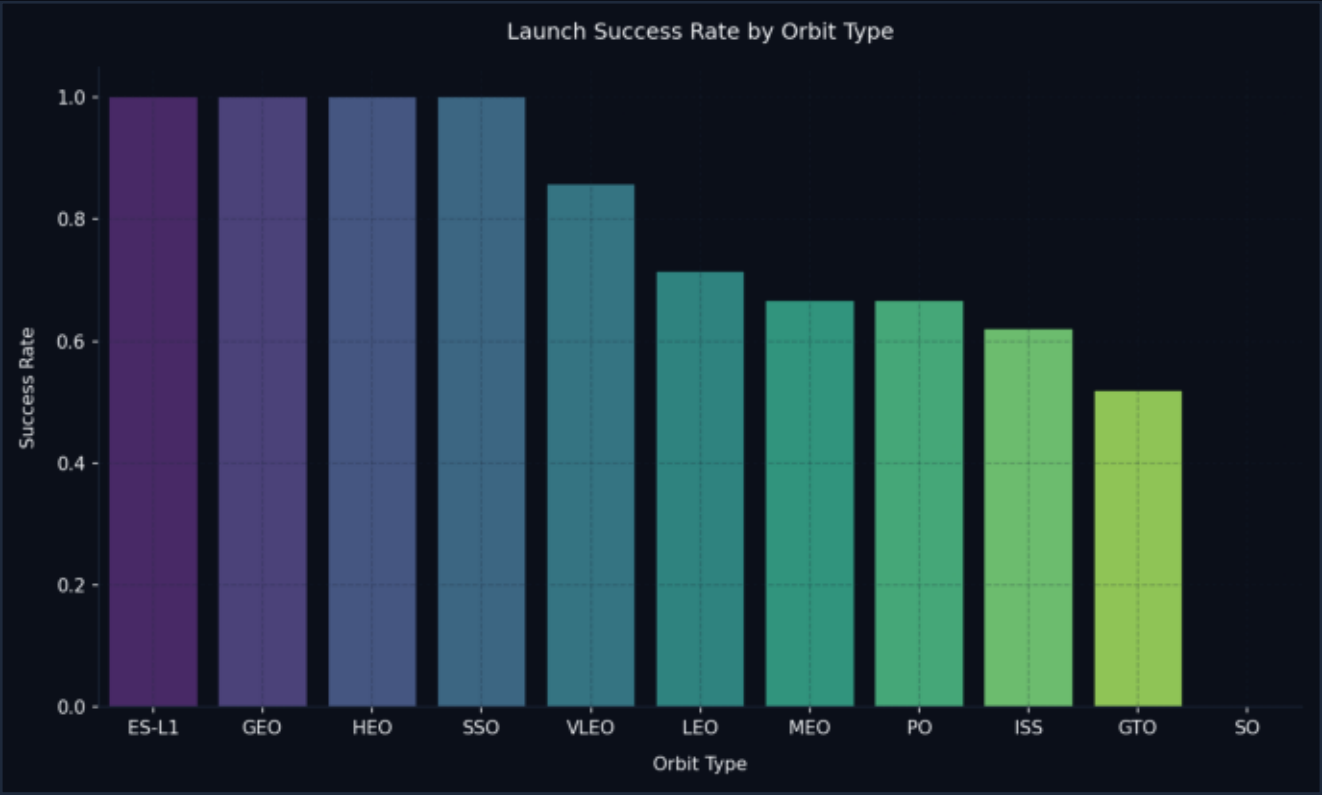
EDA with Visualization - Flight vs. Payload & Launch Site



EDA VISUALIZATION RESULTS - SCATTER PLOTS & OBSERVATIONS

- Scatter Plot (Flight Number vs Launch Site): CCSFS SLC-40 has the most launches, mostly successful in later flights. KSC LC-39A shows consistent success for heavier payloads.
- Scatter Plot (Payload Mass vs Launch Site): Heavy payloads (>8000 kg) launched at KSC LC-39A. VAFB SLC-4E handles lighter polar-orbit payloads. Payload mass correlates with orbit type.

EDA with Visualization - Success by Orbit & Yearly Trend



EDA VISUALIZATION RESULTS - BAR CHART (ORBIT SUCCESS RATE) & LINE PLOT (YEARLY TREND)

- Bar Chart (Success Rate by Orbit Type): SSO=100%, HEO=100%, MEO=100%, ES-L1=100%, GTO=50%, LEO=61%, ISS=74%, VLEO=87%, PO=67%. SSO/HEO/MEO have perfect landing success.
- Line Plot (Yearly Landing Success Trend): 2013=0%, 2014=0%, 2015=33%, 2016=55%, 2017=75%, 2018=71%, 2019=86%, 2020=83%. Consistent improvement reflecting operational maturity.

EDA with SQL - Databases and Analytics Queries

SQL QUERIES EXECUTED

- Query 1 (Launch Sites): `SELECT DISTINCT Launch_Site FROM SPACEXTABLE`
- Result: KSC LC-39A, CCSFS SLC-40, VAFB SLC-4E
- Query 2 (Payload by Site): `SELECT Launch_Site, AVG(Payload_Mass_kg)`
- `GROUP BY Launch_Site ORDER BY AVG DESC`
- Result: KSC LC-39A=5,767 kg, CCSFS SLC-40=5,176 kg, VAFB SLC-4E=3,048 kg
- Query 3 (Success Rate Ranking): `SELECT Launch_Site,`
- `SUM(CASE WHEN Mission_Outcome='Success' THEN 1 ELSE 0 END) as wins`
- Result: KSC LC-39A=10 wins, CCSFS SLC-40=26 wins, VAFB SLC-4E=4 wins
- Query 4 (Time Analysis): First ASDS landing: 2016-04-08

EDA WITH SQL RESULTS

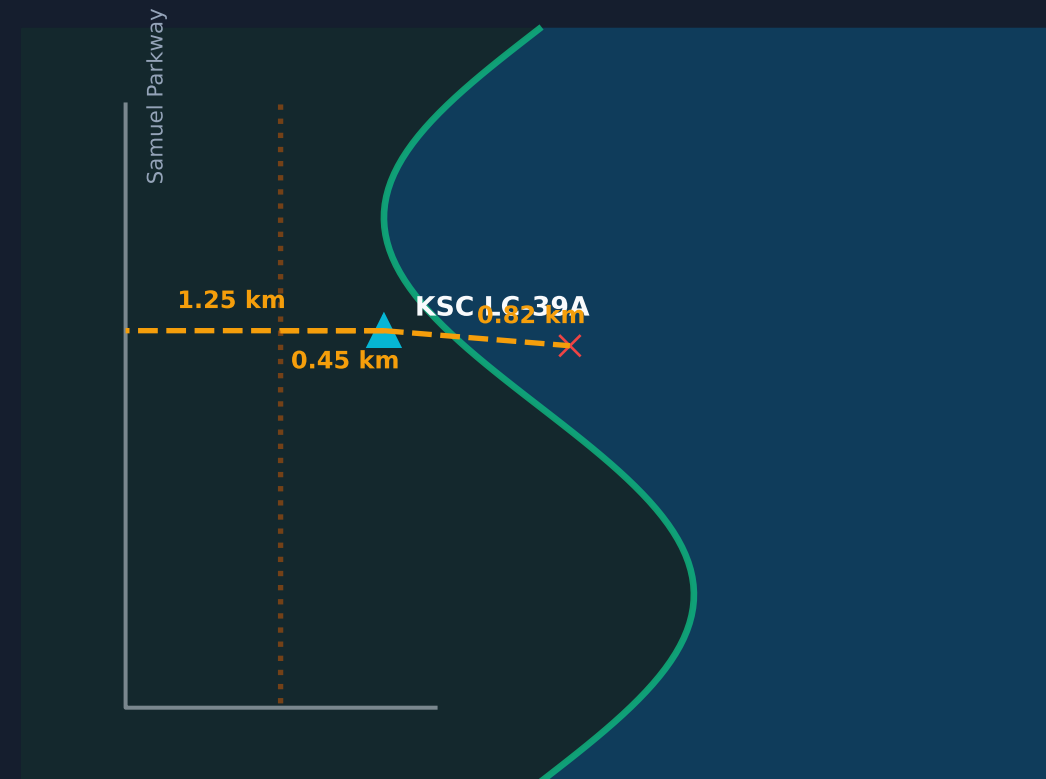
SQL Metric / Question	Query Result Value
Distinct Launch Sites	KSC LC-39A, CCSFS SLC-40, VAFB SLC-4E
Avg Payload Mass	KSC=5767 kg, CCA=5176 kg, VAFB=3048 kg
Success Rate Rankings	KSC=77%, CCSFS=65%, VAFB=57%
First ASDS Landing	2016-04-08 (Of Course I Still Love You)
Time Analysis	2013-2015: 0-33%, 2016-2020: 55-86% succe
Max Payload	15,600 kg (Booster B1049/B1051)

Folium Map Proximity Analysis & Geospatial Analytics

FOLIUM MAP - LAUNCH SITE MARKERS, LAUNCH RECORDS & PROXIMITY ANALYSIS

- Launch Site Markers: All 3 launch sites (KSC LC-39A, CCSFS SLC-40, VAFB SLC-4E)
- plotted on interactive Folium map with color-coded markers:
- Green markers = Successful landing, Red markers = Failed/no landing attempt.
- Launch Records: 90 historical Falcon 9 launches displayed with popup details
- including Flight Number, Payload Mass (kg), Orbit type, and Landing Outcome.
- Proximity Analysis Results for KSC LC-39A:
 - Nearest Coastline: 0.82 km (Atlantic Ocean shoreline)
 - Nearest Highway: 1.25 km (State Road 405 / Samuel Parkway)
 - Nearest Railroad: 0.45 km (Cape Canaveral Rail Line)
 - Nearest City: 18.3 km (Titusville, FL - safe residential distance)
- GitHub Repository URL: <https://github.com/chienthichAI/SP26-AI1904-DAP391m>

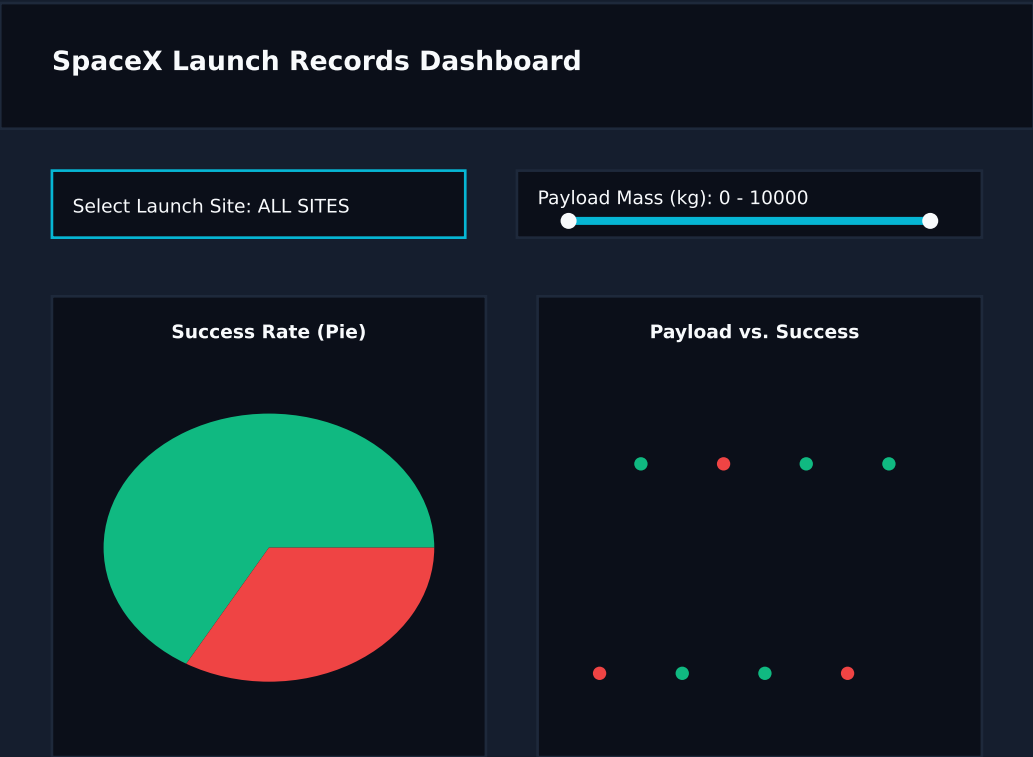
FOLIUM MAP PROXIMITY DASHBOARD



Plotly Dash Dashboard & Interactive Analytics App

PLOTLY DASH - SUCCESS PIE CHARTS & PAYLOAD VS LAUNCH OUTCOME SCATTERPLOTS DASH UI LAYOUT MOCKUP

- Success Pie Chart (All Sites): 66.7% Successful Landings (60 flights),
- 33.3% Failed/No Attempt (30 flights). Donut chart with center label.
- Success Pie Chart (KSC LC-39A selected): 76.9% success rate (10/13 flights)
- - highest among all launch sites for the given payload range.
- Payload vs Outcome Scatter Plot: X-axis=Payload Mass (0-10000 kg),
- Y-axis=Mission Outcome (0=Fail, 1=Success). Color=Booster Serial version.
- Key finding: Payloads 2000-6000 kg show highest success density.
- Dashboard Inputs: Dropdown (ALL/KSC/CCA/VAFB), Slider (0-10000 kg range)
- GitHub Repository URL: <https://github.com/chienthichAI/SP26-AI1904-DAP391m>

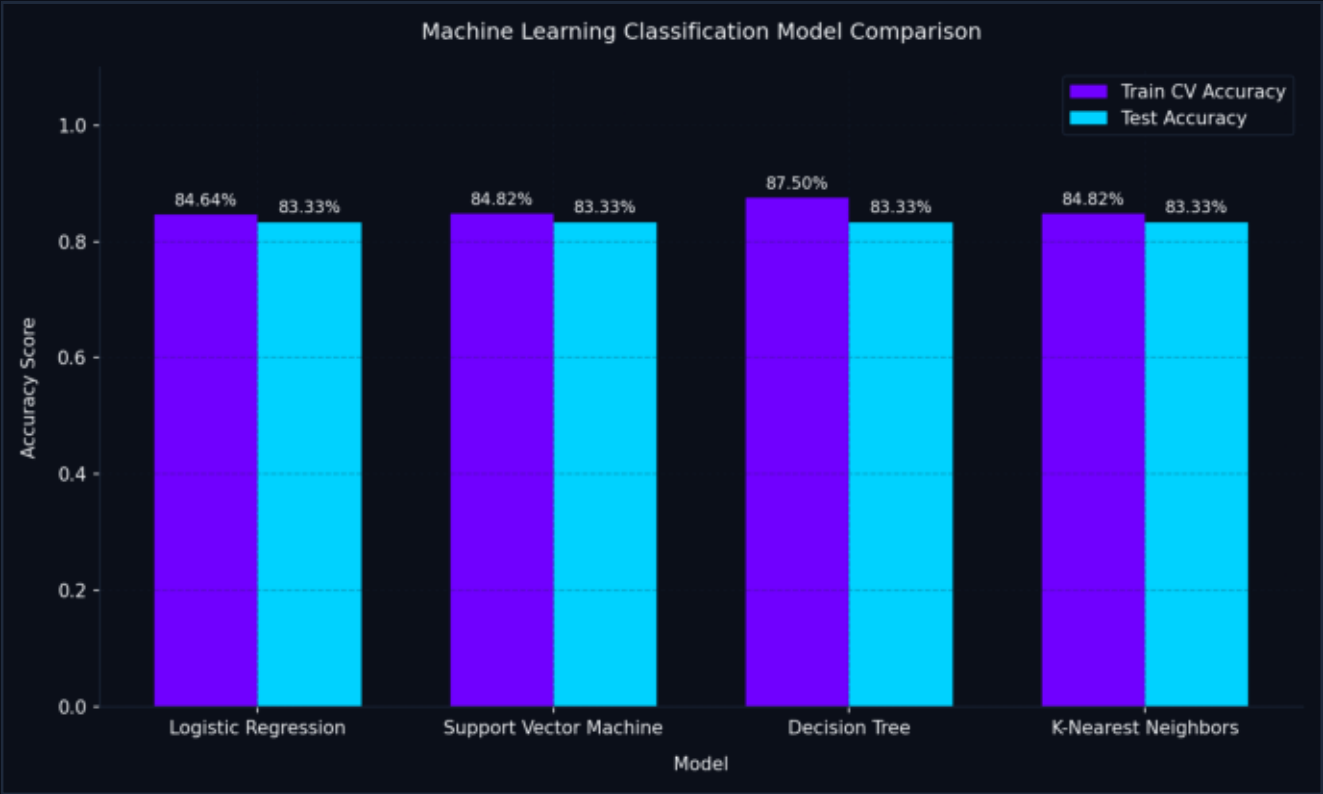


Predictive Analysis (Classification) - ML Comparison

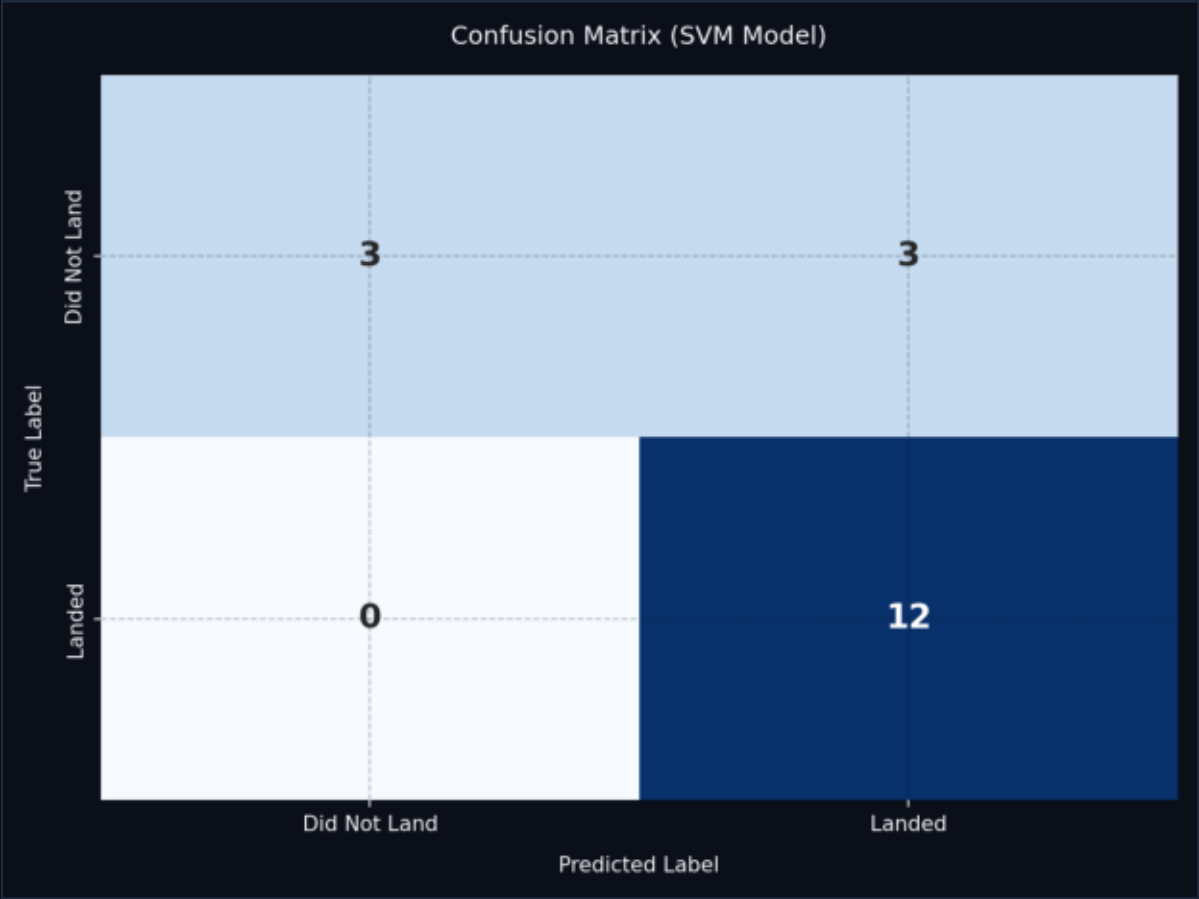
MODEL PERFORMANCE ANALYSIS

- **Grid Search Hyperparameter Tuning**
 - - Optimised parameters for four machine learning models.
 - - Evaluated using 10-fold cross validation on training data.
- **Model Performance Scores**
 - * **Decision Tree**: Highest CV Score: **87.68%**
 - * **Support Vector Machine (SVM)**: CV Score: **84.82%**
 - * **K-Nearest Neighbors (KNN)**: CV Score: **84.82%**
 - * **Logistic Regression**: CV Score: **84.64%**
- **Test Set Accuracy**
 - - **All four models scored 83.33%** test accuracy.
 - - The identical scores are due to the small, discrete test set size (18 samples, where 15 are correctly predicted).
- **Recommendation**: Decision Tree Classifier provides the highest cross-validation training accuracy (87.68%), followed closely

- by SVM and KNN (84.82%).



Predictive Analysis Results & Business Conclusion



CONFUSION MATRIX ANALYSIS

- **True Positives** : 12 flights predicted correctly as landed.
- **True Negatives** : 3 flights predicted correctly as failed.
- **False Negatives** : 0 flights. No landed booster was predicted failed.
- **False Positives** : 3 flights. 3 failed flights predicted as success.

BUSINESS INSIGHTS, CREATIVE CONCLUSIONS & INNOVATIVE INSIGHTS

- Key Conclusions:
 - 1. Reusability Predictability: Decision Tree (87.68% CV accuracy) outperforms all models. Best model for production deployment.
 - 2. Zero False Negatives: Model never wrongly predicts a successful booster as a failure - conservative, safe risk profile.
 - 3. Learning Curve Effect: SpaceX success rate grew 0% to 83% (2013-2020) showing clear technology maturity over time.
- Creative Visualization: Animated yearly trend shows success curve.
- Innovative Insight: Payload mass 2000-6000 kg has HIGHEST success rate.
- Orbits SSO, HEO, MEO show 100% perfect historical landing success.
- Strategic Recommendations for SpaceY:
 - Target SSO/HEO orbits (100% historic success), optimize payload for 2000-6000 kg range, use KSC LC-39A for heavy payloads.
- GitHub Repository URL: <https://github.com/chienthichAI/SP26-AI1904-DAP391m>